



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 1.8.2026

In addition to Part I (General Handout for all courses, appended to the Timetable), this part provides further specific details regarding the course.

Course Number	ENVS F211
Course Title	Hydraulic and Fluid Mechanics
Instructor-in-charge Lecture L1 T Th F 2 1219	Prof. SHIBANI KHANRA JHA Email: shibani@pilani.bits-pilani.ac.in Chamber No - 1210H FDI
Tutorial F9 1226	

1. Course Description:

Dimensions, units, dimensionless ratios, hydraulic similarity and the testing of models, Continuity and Bernoulli's equations, fluid properties, Reynolds experiment, laminar and turbulent flow in pipes, major and minor losses in flow through pipes, Flow in pipelines between reservoirs, open channel flow, Pumps, flow and velocity measurement devices.

2. Scope & Objective:

This course introduces the foundational principles of fluid mechanics and their practical applications in hydraulic systems used in water and wastewater treatment plants and sewerage networks. It covers the behavior of fluids in motion and at rest, emphasizing both theoretical frameworks and engineering applications. Topics include flow regimes, pressure losses, open channel flow, pump performance, and hydraulic modeling. Students will develop analytical skills to solve real-world fluid flow problems and design efficient hydraulic systems.

3. Course Learning Outcomes (CLOs):

Upon the completion of this course, students will be able to perform the following:

1. Fundamentals of Fluid Mechanics: Understand physical dimensions, units, and dimensionless analysis using Rayleigh and Buckingham π methods; Grasp the concept of hydraulic similarity and model testing
2. Fluid Properties and Flow Regimes: Analyze dynamic and kinematic viscosity; Distinguish between laminar and turbulent flow using Reynolds experiment and Reynolds number.
3. Flow in Pipes: Apply Poiseuille's law for laminar flow pressure drop; Analyze velocity profiles and pressure loss in laminar and turbulent pipe flow; Use Moody diagram and Darcy's law to determine friction factors.
4. Complex Pipe Flow Systems: Evaluate flow in non-horizontal and variable diameter pipes; Estimate minor losses due to expansions, contractions, and fittings.
5. Pipeline Flow Between Reservoirs: Assess head losses and solve problems involving series and parallel pipelines.
6. Open Channel Flow: Define hydraulic parameters like wetted perimeter and hydraulic mean depth; Use Chezy's and Manning's equations to calculate discharge and flow depth.



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7. Pumps and Hydraulic Machinery: Understand types of rotodynamic pumps and their applications; Analyze pump performance, derive affinity laws, and interpret characteristic curves; Evaluate system losses and pump operation in series and parallel configurations.

4. Textbooks and Reference Books:

T1B. P.N. Modi & S.M. Seth. Hydraulics and Fluid Mechanics Including Hydraulic Machines. 24th Ed., 2019. Standard Book House.

T2B. R.K. Bansal. Fluid Mechanics and Hydraulic Machines. 10th Ed., 2019. Laxmi Publications

T3B. Frank M. White. Fluid Mechanics. 9th Ed., 2021. McGraw-Hill

R1B. Douglas, J.F., Gasiorek, J.M., Swaffield, J.A. Fluid Mechanics 6th, Pearson/Prentice Hall, 2011 10: 0273717723

R2B. Massey B.S. and Ward-Smith J. Mechanics of Fluids. Stanley Thornes, 2012

5. Course Format: Weekly Plan (for 14 Weeks)

Week No.	Module No.	Description – Lecture Sessions	Reference	Learning Outcomes
Week 1	Module 1 Fundamentals of Fluid Mechanics	Lecture 1: Introduction to fluid mechanics, dimensions and units. Lecture 2: Fundamental and derived units; dimensional homogeneity. Lecture 3: Dimensionless ratios and their engineering significance.	Chapter: Dimensional Analysis and Hydraulic Similarity.	Explain the importance of dimensions and units, apply dimensional homogeneity, and identify commonly used dimensionless parameters.
Week 2	Module 1 Fundamentals of Fluid Mechanics	Lecture 4: Rayleigh method for dimensional analysis. Lecture 5: Buckingham π theorem and applications. Lecture 6: Hydraulic similarity, model laws, and testing of hydraulic models.	Chapter: Dimensional Analysis and Hydraulic Similarity.	Apply Rayleigh and Buckingham π methods, develop dimensionless groups, and analyze similarity between prototype and model.
Week 3	Module 2 Fluid Properties and Flow Regimes:	Lecture 7: Review of continuity equation. Lecture 8: Bernoulli's equation and applications. Lecture 9: Dynamic and kinematic viscosity; measurement of viscosity.	Chapters: Fluid Properties and Fluid Kinematics.	Apply continuity and Bernoulli equations and distinguish between dynamic and kinematic viscosity.
Week 4	Module 2 Fluid Properties and Flow Regimes	Lecture 10: Reynolds experiment and flow visualization. Lecture 11: Laminar and turbulent flow regimes; Reynolds number.	Chapter: Flow Through Pipes.	Classify flow regimes using Reynolds number and explain velocity distribution in laminar pipe flow.



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Week No.	Module No.	Description – Lecture Sessions	Reference	Learning Outcomes
		Lecture 12: Laminar flow in circular pipes; velocity profile.		
Week 5	Module 3 Flow in Pipes and Complex Pipe Flow Systems	Lecture 13: Pressure drop in laminar flow; Hagen–Poiseuille equation. Lecture 14: Development length in laminar flow. Lecture 15: Relationship between average and maximum velocities.	Chapter: Flow Through Pipes.	Calculate pressure losses in laminar flow and evaluate fully developed flow characteristics.
Week 6	Module 3 Flow in Pipes and Complex Pipe Flow Systems	Lecture 16: Turbulent flow in horizontal pipes. Lecture 17: Friction factor; Moody diagram. Lecture 18: Darcy–Weisbach equation in pressure and head forms.	Chapter: Flow Through Pipes.	Determine friction factor and estimate head and pressure losses in turbulent pipe flow.
Week 7	Module 3 Flow in Pipes and Complex Pipe Flow Systems	Lecture 19: Frictional losses in inclined pipes. Lecture 20: Flow through pipes of varying diameters. Lecture 21: Minor losses due to sudden expansion and contraction.	Chapter: Flow Through Pipes.	Analyze frictional and minor losses in practical pipeline systems.
Week 8	Module 4 Pipeline Flow Between Reservoirs	Lecture 22: Flow through pipelines between reservoirs. Lecture 23: Energy equation and relationship between losses and available head. Lecture 24: Numerical problems on reservoir-pipeline systems.	Chapter: Pipe Networks and Pipe Flow.	Analyze pipeline flow between reservoirs and estimate discharge considering head losses.
Week 9	Module 4 Pipeline Flow Between Reservoirs	Lecture 25: Branching pipelines. Lecture 26: Parallel pipeline systems. Lecture 27: Design examples and applications.	Chapter: Pipe Networks and Pipe Flow.	Analyze discharge distribution in branched and parallel pipeline networks.
Week	Module 5	Lecture 28: Introduction to open channel	Chapter:	Define open channel flow



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Week No.	Module No.	Description – Lecture Sessions	Reference	Learning Outcomes
10	Open Channel Flow	flow. Lecture 29: Wetted perimeter, hydraulic radius, hydraulic mean depth, hydraulic gradient. Lecture 30: Chezy equation and applications.	Open Channel Flow.	parameters and apply Chezy equation to determine discharge.
Week 11	Module 5: Open Channel Flow	Lecture 31: Manning's equation for discharge. Lecture 32: Determination of normal depth using Manning equation. Lecture 33: Economical channel sections and maximum discharge conditions.	Chapter: Open Channel Flow.	Compute discharge using Manning's equation and design efficient channel sections.
Week 12	Module 6: Pumps and Hydraulic Machinery	Lecture 34: Introduction to pumps and pumping systems. Lecture 35: Types of rotodynamic pumps and pump efficiencies. Lecture 36: Pump affinity laws and derivation.	Chapters: Centrifugal Pumps and Hydraulic Machines.	Classify rotodynamic pumps, evaluate pump efficiency, and apply the affinity laws to predict performance.
Week 13	Module 6 Pumps and Hydraulic Machinery	Lecture 37: Pump characteristic curves. Lecture 38: Losses in rising mains; pumps in series and parallel. Lecture 39: Course review, integrated numerical problems, and revision.	Chapters: Centrifugal Pumps and Hydraulic Machines.	Interpret pump characteristic curves, analyze pump systems in series/parallel, and solve integrated fluid flow and pumping problems.
Week 14	Revision	Doubt sessions		

6. Evaluation Schemes:

Components	Duration (Minutes)	Weightage % (Marks)	Date & Time	Nature of component (Close Book/ Open Book) Syllabus
Mid-Semester	90 min	20 (40M)	08/10/2026 AN1 AN1: 2:00 P.M. to 3:30 P.M.	CB
Comprehensive Examination	180 min	40 (80M)	12/12/2026 FN FN: 8.00 A.M. to 11.00 A.M.	CB+OB
Class Assessment	Class hour	10 (20M)	TBA	CB/OB
Tutorial	Tutorial Hour	30 (60M)	TBA	CB/OB



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Criterion for NC: [30% of the average of the top three performers or 40% of the median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.]

Remarks: No makeup is permitted for Class assessment, Tutorial and Practical classes

7. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components. The 10% class assessment is linked to attendance.
8. **Chamber Consultation Hour:** Consultation Hour – **Friday 10:00-11:00 am**. Note: IC can be reached through email or in person in the chamber.
9. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for the Mid-sem or comprehensive exam only.

Prof. Shibani Khanra Jha
Instructor-in-Charge
ENVS F21